

ABSTRACT

The rapid expansion of online advertising has boosted the digital economy but also fueled click fraud, where bots or malicious actors generate fake clicks to deplete advertiser budgets and skew metrics. This undermines ROAS and trust in advertising platforms and continues despite defenses. Machine learning (ML) and deep learning (DL) methods are extensively used to detect click fraud by analyzing user behavior, session attributes, and browsing patterns to distinguish legitimate from fraudulent interactions.

Existing approaches leverage feature engineering and classification across ML and DL models. Tree-based methods like Decision Trees (DT) and Random Forests (RF) achieve over 98% accuracy with strong interpretability, while Gradient Boosting, LightGBM, and XGBoost provide near-perfect precision and recall. Deep learning models—CNN, DNN, and RNN— capture complex sequential and behavioral patterns, with RNNs excelling in temporal features such as click intervals, scrolling, and session durations. Challenges remain in adaptability, scalability, and evolving fraud resilience.

To address these challenges, this project proposes an Adaptive Hybrid Framework for Real Time Click Fraud Detection that integrates interpretable ML models with temporal DL architectures. The framework combines feature engineering, behavioral analysis, and sequential modeling to identify subtle fraudulent patterns overlooked by traditional approaches. Tree-based classifiers efficiently detect high-risk clicks, while LSTM-based RNNs capture evolving user interactions. A dual-layer defense mechanism enables server and client-side monitoring, and incremental learning ensures continuous adaptation. Experiments show reduced false negatives, balanced efficiency, and superior performance over standalone ML or DL models.

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1. INTRODUCTION

Online advertising has become one of the most important revenue sources for businesses and digital platforms. Advertisers invest in campaigns where they pay for user interactions such as clicks, impressions, and conversions to promote their products and services. This model has played a major role in the growth of the digital economy by allowing businesses to reach a larger audience and improve their visibility in the market.

However, the rapid growth of online advertising has also led to the rise of click fraud, which has become a serious concern for advertisers and advertising platforms. Click fraud occurs when fake clicks are generated using automated bots, scripts, or coordinated human activities. These fraudulent actions are intended to misuse advertising budgets, increase click counts artificially, and manipulate campaign performance metrics. As a result, advertisers lose money and receive inaccurate data about their campaigns.

Click fraud has a significant negative impact on businesses. It reduces the return on investment (ROI) by wasting money on invalid clicks and produces misleading analytics. Due to this, companies may make incorrect decisions based on false information. Over time, this also reduces trust in digital advertising systems.

Traditional fraud detection methods are mainly based on rule-based systems and statistical techniques. These methods use predefined conditions to detect suspicious activity, such as repeated clicks from the same IP address. However, these approaches are no longer effective because fraudsters continuously change their strategies to mimic real user behavior. This makes it difficult to detect fraud using simple rules.

Machine learning techniques have improved fraud detection by analyzing patterns in user behavior and identifying unusual activities. However, these models mainly focus on structured data and may not capture time-based or sequential patterns effectively. On the other hand, deep learning models can analyze complex and temporal patterns but require more data and computational resources.

To overcome these challenges, there is a need for a system that can combine the strengths of both approaches. Such a system should be capable of analyzing both static and dynamic user behavior while also adapting to new fraud techniques.

To address this problem, this project proposes an **Adaptive Hybrid Framework for Real-Time Ad Click Fraud Detection**. The system combines machine learning and deep learning models to improve detection accuracy. Machine learning models analyze structured data such as click frequency and device information, while deep learning models like LSTM capture sequential patterns such as click intervals and user behavior over time.

By integrating these techniques, the system can detect fraudulent activities more effectively and in real time. The hybrid approach reduces errors, improves reliability, and provides a scalable solution for modern online advertising platforms.

2. LITERATURE SURVEY

Various research studies have been conducted to address the problem of click fraud in online advertising using different techniques and methodologies. In the early stages, detection approaches mainly relied on rule-based systems and statistical methods. These techniques used predefined rules such as identifying repeated clicks from the same IP address or detecting unusually high click frequencies. Although these methods were simple and easy to implement, they lacked the ability to detect complex and evolving fraud patterns. Fraudsters could easily bypass such systems by slightly modifying their behavior.

With the advancement of technology, machine learning techniques have been widely adopted for click fraud detection. Models such as Decision Trees, Random Forest, Support Vector Machines (SVM), and Logistic Regression have shown promising results by analyzing user behavior and identifying abnormal patterns. These models use features such as click frequency, session duration, device information, and browsing behavior to classify clicks as legitimate or fraudulent. Among these, tree-based models like Random Forest have gained significant attention due to their high accuracy, robustness, and ability to handle large datasets. Additionally, they provide better interpretability compared to other complex models, making them suitable for practical applications.

In addition to machine learning, deep learning techniques have also been explored to enhance fraud detection performance. Models such as Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), and Recurrent Neural Networks (RNN) are capable of capturing complex and non-linear patterns in large-scale datasets. Among these, RNN and its advanced variant LSTM (Long Short-Term Memory) have proven to be highly effective in analyzing temporal and sequential data. These models are particularly useful for understanding time-based user behavior, such as click intervals, browsing sequences, and session duration. By capturing these temporal dependencies, deep learning models can identify hidden and sophisticated fraud patterns that are not easily detected by traditional machine learning approaches.

Despite these advancements, existing approaches still face several challenges. Machine learning models are efficient in handling structured and static data but may fail to capture temporal dependencies and sequential patterns. On the other hand, deep learning models require large volumes of labeled data, high computational resources, and longer training times. Additionally, many existing systems lack adaptability and are not capable of detecting new and evolving fraud techniques in real time.

To overcome these limitations, recent research has focused on hybrid approaches that combine machine learning and deep learning models. These hybrid systems leverage the strengths of both techniques, where machine learning models efficiently analyze structured features and deep learning models capture temporal and behavioral patterns. This combination improves detection accuracy, reduces false positives and false negatives, and enhances the overall performance of the system.

Furthermore, modern studies emphasize the importance of real-time fraud detection, scalability, and adaptive learning. Systems are being designed to continuously learn from new data and update their models

to handle evolving fraud patterns. These advancements aim to make fraud detection systems more robust, intelligent, and efficient in real-world applications.

Based on these research findings, the proposed system adopts an adaptive hybrid framework that integrates Random Forest and LSTM models. This approach combines the advantages of both machine learning and deep learning techniques to achieve more accurate, scalable, and real-time click fraud detection.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing click fraud detection systems mainly rely on traditional techniques such as rule-based filtering methods, statistical threshold techniques, machine learning models, and deep learning models. Rule-based systems operate using predefined conditions to identify fraudulent activities. For example, clicks from the same IP address occurring repeatedly within a short time or unusually high click frequency are marked as suspicious. Although these methods are simple and easy to implement, they are not effective in handling complex and evolving fraud patterns.

Statistical threshold techniques are also commonly used, where fixed limits are defined for parameters such as click rate, session duration, and time intervals between clicks. If these values exceed predefined thresholds, the system flags the activity as fraudulent. However, these approaches lack flexibility and fail to adapt to changing fraud behaviors, as fraudsters can easily modify their patterns to stay within thresholds. To improve detection accuracy, machine learning models such as Decision Tree and Random Forest have been introduced. These models analyze structured data and identify patterns in user behavior to classify clicks as legitimate or fraudulent. Similarly, deep learning models such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) are used to detect complex patterns in large datasets.

Despite these advancements, most existing systems rely heavily on static feature-based classification. They mainly focus on parameters such as IP address frequency, click rate, device information, browser details, and session duration. These systems fail to capture dynamic and temporal user behavior, which is essential for identifying advanced fraud techniques like bot-driven and automated click patterns.

3.2 DISADVANTAGES OF EXISTING SYSTEMS

The existing systems suffer from several major limitations. One of the key challenges is their inability to detect new and evolving fraud patterns. Fraudsters continuously develop advanced techniques to bypass detection systems, making traditional rule-based and static approaches ineffective.

Another significant issue is the high rate of false positives and false negatives. Legitimate user clicks may be incorrectly classified as fraudulent, while actual fraudulent clicks may go undetected. This reduces the reliability and accuracy of the system.

Most existing systems do not support real-time detection. They process data after it has been collected, which delays fraud identification and allows malicious activities to continue for longer periods. This delay can result in increased financial losses.

Additionally, these systems provide limited analysis of behavioral and temporal patterns. They do not effectively capture sequential activities such as click intervals, browsing behavior, and user interaction patterns, which are critical indicators of fraud.

Scalability is another major limitation. With the rapid growth of online advertising data, existing systems struggle to handle large datasets efficiently. Performance degradation occurs when the system processes high volumes of data.

Finally, the dependence on manually defined rules makes these systems less flexible and difficult to maintain. Any changes in fraud patterns require manual updates, which is time-consuming and inefficient.

3.3 PROPOSED SYSTEM

To overcome the limitations of existing systems, the proposed system introduces an **Adaptive Hybrid Framework for Real-Time Ad Click Fraud Detection**. This framework combines machine learning and deep learning techniques to improve detection accuracy, efficiency, and adaptability.

The system is designed to analyze both static and dynamic user behavior. Machine learning models such as Random Forest are used to process structured data and detect suspicious patterns based on features like click duration, device type, browser information, click frequency, and session details. These models are efficient, fast, and capable of handling large datasets.

At the same time, deep learning models such as Long Short-Term Memory (LSTM) networks are used to capture sequential and temporal patterns in user behavior. LSTM models analyze time-based data such as click intervals, browsing sequences, and repeated user actions. This allows the system to identify complex fraud behaviors, including automated bot activities, which cannot be detected using traditional methods.

The outputs from both machine learning and deep learning models are combined using a hybrid approach in the decision layer. This combination leverages the strengths of both models, improving overall system performance. The hybrid model significantly reduces false positives and false negatives, ensuring more accurate and reliable predictions.

The system also supports real-time fraud detection, enabling immediate identification of suspicious activities. Additionally, it incorporates incremental learning, where new data is continuously added and used to update the models. This allows the system to adapt to evolving fraud patterns and maintain high detection accuracy over time.

3.4 ADVANTAGES OF PROPOSED SYSTEM

The proposed system offers several advantages over traditional methods. It provides improved detection accuracy by combining machine learning and deep learning models. The hybrid approach ensures better performance by analyzing both structured data and sequential user behavior, enabling the system to detect a wider range of fraud patterns.

The system reduces false positives and false negatives, resulting in more reliable and accurate predictions. It supports real-time detection, allowing quick identification and prevention of fraudulent activities, which helps in minimizing financial losses for advertisers.

The ability to analyze both behavioral and temporal patterns makes the system highly effective in detecting complex fraud techniques such as bot-generated clicks and automated scripts. Unlike traditional systems, it can identify subtle and hidden patterns that evolve over time.

The system is highly scalable and capable of handling large volumes of data efficiently, making it suitable for real-world applications and large-scale online advertising platforms. It can process continuous streams of data without significant performance degradation.

Additionally, the system supports adaptive learning through continuous dataset updates. This enables it to respond to new and evolving fraud patterns, making it more robust and future-ready. The incremental learning capability ensures that the model improves over time without requiring complete retraining.

Another important advantage is the flexibility of the system. Since it is based on a modular architecture, individual components such as models or preprocessing techniques can be updated or replaced without affecting the entire system.

The system also provides better decision-making support by generating fraud probability scores and risk levels, which help advertisers understand the severity of fraudulent activities. This improves transparency and trust in the system.

Furthermore, the integration of both machine learning and deep learning techniques ensures a balanced approach between speed and accuracy. Machine learning provides faster predictions, while deep learning captures complex patterns, resulting in an efficient and reliable system.

Overall, the proposed system is a comprehensive, scalable, and adaptive solution that significantly enhances the effectiveness of click fraud detection in modern digital advertising environments.

3.5 LIMITATIONS OF ADVANCED SYSTEM

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4.SYSTEM STUDY

4.1 FEASIBILITY STUDY

A feasibility study is a critical phase in system development that evaluates whether the proposed system is practical, efficient, and beneficial for implementation. It helps in assessing whether the system can be successfully developed within the available resources, technological capabilities, and time constraints. It also ensures that the system meets user requirements and aligns with real-world operational conditions.

For the proposed click fraud detection system, feasibility is analyzed from multiple perspectives, including economical, technical, social, operational, and practical feasibility. This comprehensive evaluation ensures that the system is not only implementable but also sustainable in the long run.

4.1.1 Economical Feasibility

Economical feasibility evaluates whether the system is cost-effective and financially viable. The proposed system uses open-source tools such as Python, Django, Scikit-learn, TensorFlow, and supporting libraries, which significantly reduce development costs.

The hardware requirements are minimal, and the system can operate on standard computing devices without requiring specialized infrastructure. This reduces both initial investment and deployment costs. Additionally, the use of automated fraud detection minimizes manual effort, thereby reducing operational and maintenance expenses.

The economic benefits of the system are substantial. It helps advertisers prevent financial losses caused by fraudulent clicks, ensures accurate campaign performance metrics, and improves return on investment (ROI). It also enhances resource utilization by eliminating unnecessary spending on invalid traffic.

In the long term, the system provides a cost-efficient solution for digital advertising platforms, making it economically sustainable and beneficial.

4.1.2 Technical Feasibility

Technical feasibility focuses on evaluating whether the system can be developed using the available technologies and technical expertise. The proposed system is built using modern and widely accepted technologies such as Python and Django for application development.

Machine learning is implemented using Scikit-learn, and deep learning models such as LSTM are developed using TensorFlow and Keras. These technologies are capable of handling large datasets, performing complex computations, and supporting real-time data processing.

The system architecture is modular, allowing different components such as data preprocessing, model prediction, and decision-making to function independently. This improves flexibility, maintainability, and scalability.

The integration of machine learning and deep learning techniques enhances the system's capability to analyze both structured and sequential data. Additionally, the availability of extensive documentation and community support makes development and troubleshooting easier.

4.1.3 Social Feasibility

Social feasibility evaluates how well the system is accepted by users and how beneficial it is to society. The proposed click fraud detection system plays an important role in improving fairness and transparency in online advertising.

By detecting fraudulent clicks, the system helps businesses make accurate decisions based on reliable data. It also builds trust between advertisers and advertising platforms by ensuring that marketing budgets are used effectively.

The system is designed to be user-friendly and easy to operate, even for users with limited technical knowledge. This increases user acceptance and usability. Additionally, it contributes to reducing unethical practices in digital advertising, promoting a more secure and reliable digital environment.

4.1.4 Operational Feasibility

Operational feasibility evaluates how effectively the system performs in real-world conditions and how easily it can be used by end users. The proposed system is designed to support real-time fraud detection, allowing immediate identification of suspicious activities and enabling quick decision-making.

The system operates automatically with minimal human intervention, which reduces operational complexity and increases efficiency. Most processes, including data preprocessing, model prediction, and fraud classification, are handled internally by the system, minimizing manual effort and reducing the chances of human error.

The system can be easily integrated into existing online advertising platforms and workflows without requiring major structural changes. This makes it practical for deployment in real-world environments. Additionally, the system is designed with a user-friendly interface that allows users to input data and view results without requiring advanced technical knowledge.

Regular monitoring and periodic updates ensure that the system remains effective over time. The system can adapt to changing fraud patterns through incremental learning and model updates. Furthermore, its ability to provide real-time responses makes it suitable for environments where immediate action is required.

4.1.5 Scalability Feasibility

Scalability feasibility evaluates the system's ability to handle increasing amounts of data, users, and processing demands without significant performance degradation. With the continuous growth of digital advertising, the volume of clickstream data is rapidly increasing, making scalability a critical requirement.

The proposed system is designed to process large datasets efficiently by utilizing optimized machine learning and deep learning models. It can handle high volumes of incoming data while maintaining performance and accuracy. The modular architecture of the system allows individual components to be scaled independently based on system requirements.

The system can also be extended to support distributed computing environments and cloud-based deployment. By leveraging cloud infrastructure, the system can dynamically allocate resources such as storage and processing power, ensuring smooth operation even under heavy workloads.

Additionally, the system can be integrated with big data technologies and real-time streaming frameworks to process continuous data streams. This makes it suitable for large-scale applications involving millions of users and high-frequency click events.

4.1.6 Security Feasibility

Security feasibility evaluates the system's ability to ensure data protection, prevent unauthorized access, and maintain the integrity of operations. The proposed system is designed to securely handle user data, click information, and prediction results.

The system can incorporate authentication and authorization mechanisms to ensure that only authorized users can access sensitive information. Data encryption techniques can be used to protect data during transmission and storage, preventing data breaches and unauthorized access.

Additionally, secure database management practices can be implemented to ensure data integrity and consistency. Regular monitoring and logging mechanisms help in detecting unusual activities and potential security threats.

The system also contributes to enhancing overall security in online advertising by detecting and preventing fraudulent activities. By identifying bot-generated clicks and malicious behavior, the system helps protect advertising platforms from financial and operational risks.

Furthermore, the system can be extended to comply with data protection regulations and industry standards, ensuring safe and ethical handling of user information.

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

The following hardware components are required to develop and run the click fraud detection system. Each component plays a specific role in the execution of the project:

- **Processor: Intel i3 or higher**

The processor is used to execute the application, handle backend processing in Django, and run machine learning and deep learning models. During prediction, it processes input data and generates results efficiently.

- **RAM: Minimum 4 GB (8 GB recommended)**

RAM is used to load datasets, run preprocessing operations, and execute models like Random Forest and LSTM. Higher RAM ensures smooth performance during model training and real-time prediction.

- **Hard Disk: Minimum 20 GB free space**

Storage is used to save datasets, trained models (such as .pkl and .keras files), project code, and system logs. It also stores prediction results in the database.

- **Operating System: Windows / Linux / macOS**

The operating system provides the environment to run Python, Django, and all required libraries. It supports development, testing, and deployment of the system.

- **Internet Connection**

The internet is used to install required libraries, update frameworks, and support web-based access to the system. It can also be used if the system is deployed online.

- **Optional GPU**

A GPU is used during the training phase of the LSTM model to speed up computations. It helps in handling large datasets and reduces training time significantly.

- **Input/Output Devices**

Input devices like keyboard and mouse are used for entering data and interacting with the system, while output devices display prediction results and system outputs.

5.2 SOFTWARE REQUIREMENTS

The following software tools and technologies are used in the development and implementation of the system:

- **Python**

Python is used as the core programming language to implement the entire system, including data preprocessing, model development, and backend logic.

- **Django Framework**

Django is used to develop the web application. It handles user input, processes requests, connects with machine learning models, and displays prediction results through the interface.

- **Scikit-learn**

Scikit-learn is used to implement the machine learning model, specifically Random Forest. It is used for training the model and making predictions based on structured input data.

- **TensorFlow / Keras**

TensorFlow and Keras are used to develop and train the LSTM model. This model analyzes sequential and time-based user behavior to detect complex fraud patterns.

- **Pandas and NumPy**

Pandas is used to read and preprocess datasets, such as cleaning and organizing data. NumPy is used for numerical operations required during model training and prediction.

- **Jupyter Notebook**

Jupyter Notebook is used during the development phase for training models, testing algorithms, and visualizing data before integrating them into the main system.

- **Visual Studio Code / PyCharm**

These IDEs are used for writing, debugging, and managing the project code. They help in efficient development and error handling.

- **Database (SQLite)**

SQLite is used to store user inputs, prediction results, and system logs. It helps in maintaining records for future analysis and system improvement.

- **Model Storage (Joblib / Pickle)**

These tools are used to save trained machine learning models so that they can be loaded during prediction without retraining.

- **Operating System**

The operating system supports all software tools and ensures smooth execution of the application.

6.SOFTWARE REQUIREMENTS SPECIFICATION

The Software Requirements Specification (SRS) provides a detailed description of the functional and non-functional requirements of the proposed click fraud detection system. It serves as a blueprint for system design, development, and implementation, ensuring that all system components meet the required objectives and performance standards.

The primary purpose of this system is to detect fraudulent clicks in online advertising using an adaptive hybrid approach that combines machine learning and deep learning techniques. The system aims to improve detection accuracy, minimize financial losses caused by invalid clicks, and provide reliable and transparent analytics for advertisers and digital platforms.

The system is designed to analyze user click data and classify each click as either legitimate or fraudulent. It uses machine learning models such as Random Forest to analyze structured and statistical features, including click frequency, device information, and session details. At the same time, deep learning models such as LSTM are used to capture sequential and temporal patterns in user behavior, such as click intervals and browsing sequences. By combining both approaches, the system can effectively detect both simple and complex fraud patterns.

The overall architecture of the system follows a layered approach. The process begins with user input, where click-related data is provided through a web interface. This data is then passed through a preprocessing layer, where it is cleaned, encoded, and scaled. After preprocessing, the data is sent to both machine learning and deep learning models for analysis. The outputs from these models are combined in a hybrid decision layer to generate the final prediction. The results are displayed to the user and stored in a database for future reference and analysis.

The system supports different types of users, including administrators and general users. Administrators are responsible for managing datasets, monitoring system performance, and updating models. Users or advertisers interact with the system by providing input data and viewing prediction results. The system is designed to be user-friendly, ensuring easy interaction without requiring advanced technical knowledge.

The functional requirements of the system include accepting user input, performing data preprocessing, extracting relevant features, applying machine learning and deep learning models, combining model outputs using a hybrid approach, generating predictions, and storing results in the database. The system also provides reporting capabilities, allowing users to analyze prediction outcomes and system performance.

In addition to functional requirements, the system also meets several non-functional requirements. It is designed to provide real-time performance, ensuring quick response and immediate fraud detection. The system maintains high accuracy and reliability by reducing false positives and false negatives. It is scalable, allowing it to handle large volumes of data efficiently. The system also ensures usability through a simple and intuitive interface. Security is a critical aspect, and appropriate measures are implemented to protect user data and prevent unauthorized access.

The system operates with basic hardware requirements such as an Intel i3 processor, a minimum of 4 GB RAM, and sufficient storage space. It is developed using software technologies such as Python for programming, Django for web development, Scikit-learn for machine learning, TensorFlow/Keras for deep learning, and SQLite for database management.

Additionally, the system supports extensibility and maintainability. New models, features, or modules can be added without affecting the overall system functionality due to its modular architecture. The system also supports incremental learning, allowing it to adapt to new and evolving fraud patterns over time.

Overall, the Software Requirements Specification outlines all necessary requirements to design, develop, and deploy an efficient, scalable, secure, and reliable click fraud detection system capable of handling real-world challenges in digital advertising.

7.ARCHITECTURE

The proposed system follows a layered architecture that integrates web technologies with machine learning and deep learning components to provide accurate and real-time click fraud detection. The architecture is designed to handle user input, process data, perform predictions, and store results efficiently.

The system begins with the user interface layer, where users interact with the application through a web-based interface developed using the Django framework. Users can input click-related data such as device information, click duration, and session details. The Django backend processes these inputs and forwards them to the data processing module.

In the data preprocessing layer, the input data is cleaned, encoded, and scaled to make it suitable for model prediction. This includes converting categorical data into numerical form and normalizing the input features.

The machine learning layer uses models such as Random Forest to analyze structured and statistical features of the data. It quickly identifies suspicious patterns based on parameters like click frequency, device type, and session behavior.

The deep learning layer uses an LSTM (Long Short-Term Memory) model to analyze sequential and temporal patterns in user behavior. It captures time-based dependencies such as repeated clicks and irregular intervals, which are strong indicators of fraudulent activity.

The outputs from both machine learning and deep learning models are combined in the hybrid decision layer. This layer merges the predictions to produce a final fraud probability, improving accuracy and reducing false detections.

The prediction results are then sent back to the user through the web interface, displaying whether the click is legitimate or fraudulent along with risk information. At the same time, the system stores the data and prediction results in the database for future reference and incremental learning.

The architecture also supports real-time processing and continuous learning, allowing the system to adapt to new fraud patterns and improve performance over time. This layered design ensures scalability, flexibility, and efficient fraud detection.

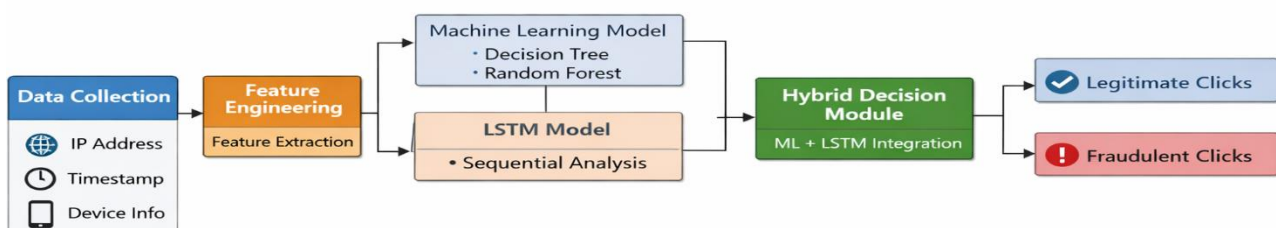


Fig: 7.1 System Architecture

System Architecture Diagram

The given diagram represents the layered architecture of the Adaptive Hybrid Click Fraud Detection System, which processes user click data through multiple stages to accurately classify clicks as legitimate or fraudulent.

Data Collection Layer

The process begins with the Data Collection module, where raw click-related data is gathered. This includes:

IP Address

Timestamp

Device Information

This data represents the user's interaction with online advertisements. It serves as the input for the entire system. The quality and completeness of this data are crucial, as they directly impact the accuracy of fraud detection.

Feature Engineering Layer

After data collection, the system moves to the Feature Engineering module, where raw data is transformed into meaningful features.

Extracts important attributes like:

Click frequency

Time intervals between clicks

Session behavior

Converts raw inputs into structured format

This step enhances the quality of input data and helps models detect hidden patterns more effectively.

Machine Learning Layer

The processed data is then passed to the Machine Learning Model, which includes:

Decision Tree

Random Forest

This layer focuses on analyzing structured and statistical features such as:

Click count

Device patterns

User behavior

It quickly identifies suspicious activities using learned patterns from historical data.

Deep Learning Layer (LSTM)

Parallel to the ML model, the data is also processed by the LSTM Model, which performs:

Sequential Analysis

Time-based behavior analysis

This layer captures:

Click intervals

Repeated sequences

Temporal dependencies

It is especially useful for detecting bot behavior, which often follows predictable patterns over time.

Hybrid Decision Module (Core Component)

The outputs from both the Machine Learning and LSTM models are sent to the Hybrid Decision Module.

Combines:

ML prediction

LSTM prediction

Uses a fusion approach (such as weighted combination) to generate the final decision.

Purpose:

Improve accuracy

Reduce false positives

Reduce false negatives

This is the most important part of the system because it leverages the strengths of both models.

Output Layer

After processing, the system classifies the result into:

Legitimate Clicks

Fraudulent Clicks

The result is displayed to the user through the web interface along with possible risk information.

Database / Storage (Implicit Layer)

Although not explicitly shown in the diagram, the system stores:

Input data

Prediction results

This data is used for:

Future analysis

Model improvement

Incremental learning

8. MODULES

The proposed click fraud detection system is designed using a modular approach, where each module performs a specific function in the overall workflow. This modular design improves system maintainability, scalability, and performance. Each module interacts with other modules to process data, perform predictions, and generate results efficiently.

User Interface Module

The User Interface Module provides a platform for users to interact with the system. It is developed using the Django web framework and offers a user-friendly environment for entering and viewing data.

This module allows users to input click-related information such as device type, browser details, click duration, and session attributes. It also displays the prediction results generated by the system, including whether a click is fraudulent or legitimate, along with the associated risk level.

The interface ensures smooth communication between the user and the backend system. It also helps in visualizing outputs clearly, making the system easy to use even for non-technical users.

Data Collection Module

The Data Collection Module is responsible for gathering input data required for fraud detection. The data can be collected from user inputs, logs, or external sources.

This module captures important parameters such as IP address, device type, browser information, click frequency, session duration, and user interaction patterns. These features are essential for identifying fraudulent activities.

The accuracy of this module directly affects the performance of the system, as high-quality data leads to better predictions. It ensures that all necessary information is collected in a structured format for further processing.

Data Preprocessing Module

The Data Preprocessing Module prepares the collected data for model prediction. Raw data often contains inconsistencies, missing values, and categorical features that need to be transformed.

This module performs several operations such as data cleaning, handling missing values, encoding categorical variables into numerical form, and feature scaling. These steps ensure that the data is compatible with machine learning and deep learning models.

Proper preprocessing improves model performance and helps in generating accurate predictions. It also reduces noise in the data and standardizes input features.

Machine Learning Module

The Machine Learning Module uses algorithms such as Random Forest to analyse structured data. It focuses on detecting patterns in statistical and behavioural features.

Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve prediction accuracy. It is effective in handling large datasets and provides quick predictions.

This module identifies suspicious activities based on features like click frequency, device type, and session behavior. It plays a crucial role in detecting fraud at an early stage using structured data analysis.

Deep Learning Module

The Deep Learning Module uses an LSTM (Long Short-Term Memory) network to analyze sequential and temporal data. Unlike traditional models, LSTM can capture time-based dependencies and patterns in user behavior.

This module detects fraud patterns such as repeated click sequences, abnormal time intervals between clicks, and automated bot activities. It is especially useful for identifying complex fraud techniques that evolve over time. The LSTM model enhances the system's ability to detect hidden patterns and improves the overall detection capability.

Hybrid Decision Module

The Hybrid Decision Module is the core component of the system. It combines the outputs from both the Machine Learning and Deep Learning modules to generate the final prediction.

The outputs from the Random Forest and LSTM models are merged using a weighted combination approach. This hybrid method leverages the strengths of both models, improving accuracy and reducing false positives and false negatives. By integrating both static and sequential analysis, this module provides a more reliable and robust fraud detection mechanism.

Prediction Module

The Prediction Module generates the final result based on the hybrid decision. It classifies each click as either fraudulent or legitimate.

In addition to classification, the module also provides a fraud probability score and risk level, helping users understand the severity of the detected activity. The results are then displayed through the user interface.

This module ensures that the output is clear, accurate, and useful for decision-making.

Database Module

The Database Module is responsible for storing all system data, including user inputs, prediction results, and system logs. It uses a database system such as SQLite to manage data efficiently.

The stored data can be used for future analysis, reporting, and improving model performance. This module ensures data integrity, consistency, and easy retrieval of information. It also supports incremental learning by storing new data for continuous system improvement.

Incremental Learning Module

The Incremental Learning Module enables the system to continuously learn from new data. Instead of relying only on static datasets, the system updates its dataset with newly observed click events. This allows the model to adapt to evolving fraud patterns and improve its performance over time. Incremental learning makes the system more dynamic and responsive to real-world changes. This module ensures that the system remains effective even as fraud techniques become more advanced.

9.DIAGRAMS

9.1 DATA FLOW DIAGRAM

The Data Flow Diagram (DFD) represents the flow of data within the click fraud detection system. It illustrates how input data is processed through different stages to generate the final output. The process begins with the **User Input**, where click-related data such as device information, click duration, and session details are provided to the system. This data is then passed to the **Data Preprocessing** stage, where it is cleaned, encoded, and scaled to make it suitable for model analysis. After preprocessing, the data is forwarded to the **Machine Learning and Deep Learning Models**. The machine learning model (Random Forest) analyses structured and statistical features, while the deep learning model (LSTM) captures sequential and temporal patterns in user behavior.

The outputs from both models are then sent to the **Hybrid Decision** stage, where they are combined using a weighted approach to generate the final fraud prediction. This improves accuracy and reduces false detections.

The result is then displayed in the **Output** stage, indicating whether the click is fraudulent or legitimate along with the risk level. Finally, the processed data and prediction results are stored in the **Database** for future reference and incremental learning.

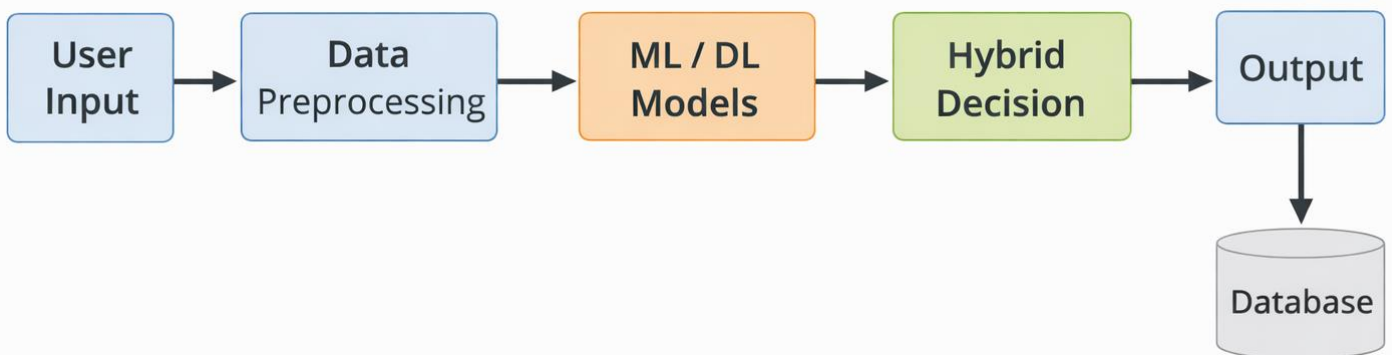


Fig: 9.1 Data Flow Diagram

9.2 USE CASE DIAGRAM

The Use Case Diagram represents the interaction between different users and the click fraud detection system. It shows how the **Admin** and **Advertiser** interact with the system to perform various operations.

The system has two main actors: **Admin** and **Advertiser**.

The **admin** plays a major role in managing and maintaining the system. The admin can **monitor traffic**, which involves observing user activity and identifying unusual patterns. The admin can also **upload click data**, which is used as input for training and analysis. Another important function is **updating the model**, where the admin retrains or improves the machine learning and deep learning models to enhance detection

accuracy. The admin can also **view reports**, which provide detailed insights about fraud detection results and system performance.

The **Advertiser** interacts with the system mainly to check outcomes. The advertiser can **view reports** to analyse campaign performance and can also **view fraud status**, which shows whether clicks are legitimate or fraudulent.

The use case **Detect Fraud** is a core function of the system, which is triggered by actions such as monitoring traffic or uploading click data. This process uses machine learning and deep learning models to analyse data and identify fraudulent activities.

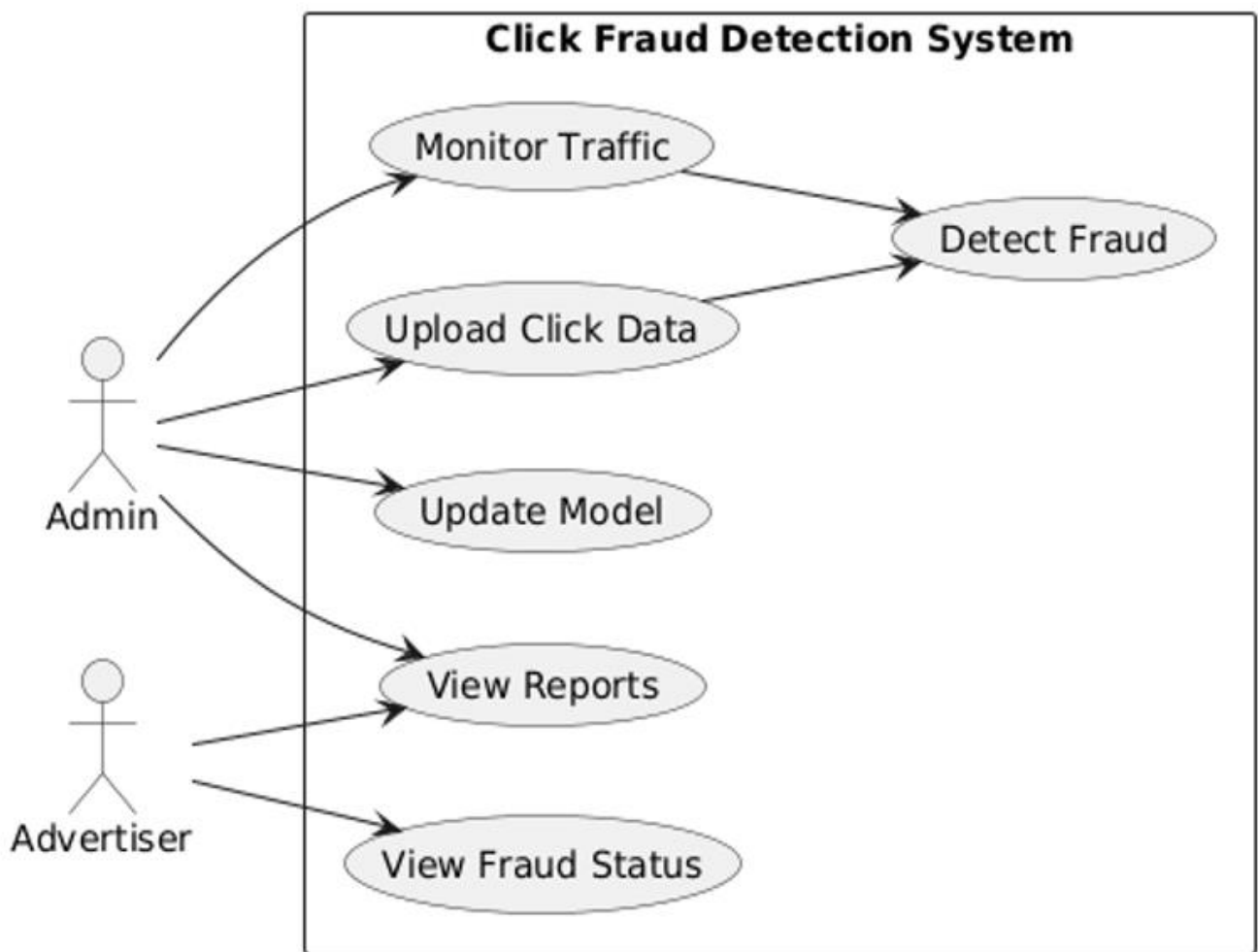


Fig: 9.2 Use Case Diagram

9.3 CLASS DIAGRAM

The class diagram represents the structure of the click fraud detection system by showing different classes, their attributes, and methods, as well as the relationships between them.

The **ClickData** class is responsible for storing input data related to user clicks. It contains attributes such as IP address, device information, timestamp, and click count. This data acts as the primary input for the system.

The **FeatureExtractor** class processes the raw click data and extracts relevant features required for analysis. It uses the method `extractFeatures()` to convert raw data into a structured format suitable for machine learning and deep learning models.

The processed data is then passed to two important classes: **MLModel** and **LSTMModel**. The **MLModel** class uses machine learning techniques to predict whether a click is fraudulent or not using the `predict()` method. The **LSTMModel** class analyzes sequential and temporal patterns in user behavior using the `analyzeSequence()` method. These two models work independently to generate predictions.

The outputs from both models are then sent to the **FraudDetector** class, which acts as the core component of the system. It combines the results from the machine learning and deep learning models using the `combineResults()` method and makes the final fraud decision using the `detectFraud()` method.

The **ReportGenerator** class is responsible for generating reports based on the detection results. It uses the `generateReport()` method to provide meaningful insights and summaries.

The **User** class represents the system user, who can access the results through the `viewReports()` method.

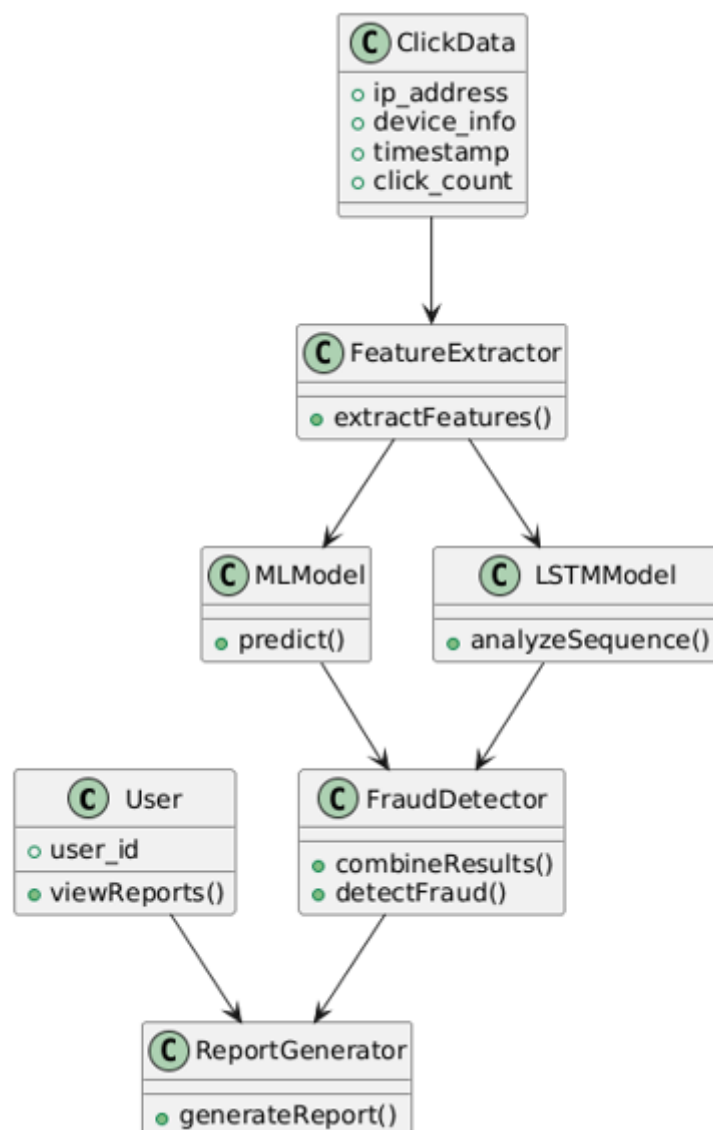


Fig: 9.3 Class Diagram

9.4 SEQUENCE DIAGRAM

The sequence diagram represents the step-by-step interaction between different components of the click fraud detection system over time. It shows how data flows from the user to various modules and how the final result is generated. The process begins when the **User uploads click data** to the system. This data includes information such as device details, click duration, and session attributes. The **System** receives this input and forwards it to the **Feature Extractor** module.

The **Feature Extractor** processes the input data by performing preprocessing and extracting relevant features required for analysis. These features are then sent to both the **Machine Learning Model** and the **LSTM Model**.

The **Machine Learning Model** analyses the structured features and generates a prediction based on statistical patterns. At the same time, the **LSTM Model** processes the sequential data and identifies temporal patterns in user behaviour, producing its own prediction.

Both predictions are then sent to the **Fraud Detector**, which acts as the core component of the system. It combines the outputs from the machine learning and deep learning models to make the final decision using a hybrid approach.

Once the final decision is made, the result is sent back to the **System**, which then displays the output to the user. The output indicates whether the click is **fraudulent or legitimate**.

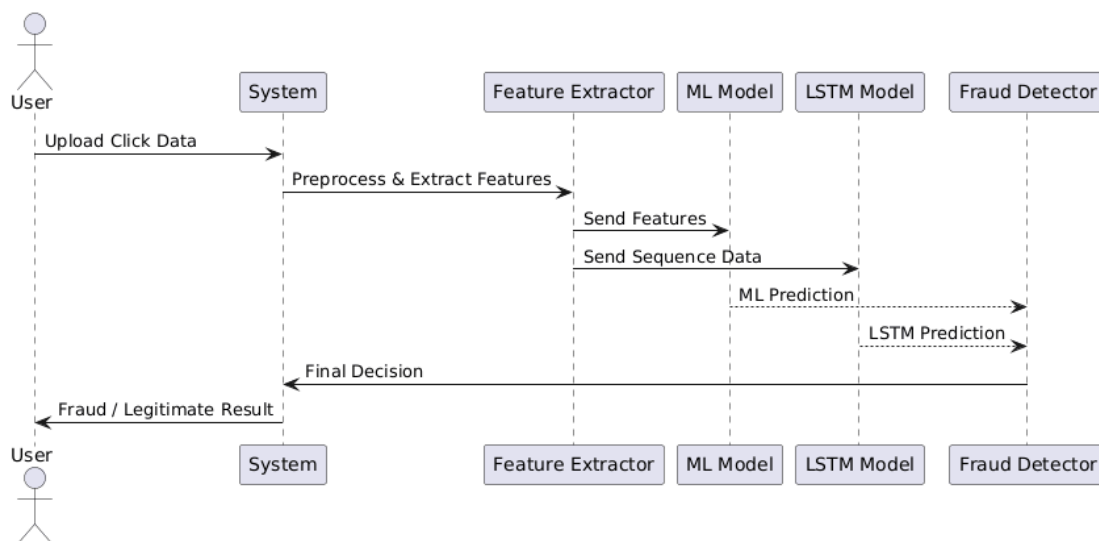


Fig 9.4 Sequences Diagram

9.5 ACTIVITY DIAGRAM

The activity diagram represents the workflow of the click fraud detection system, showing the sequence of operations from data input to final result storage.

The process begins with the **collection of click data**, where user interaction details such as device information, click duration, and session attributes are gathered. This data is then passed to the **data preprocessing** stage, where it is cleaned, encoded, and prepared for further analysis.

Next, the system performs **feature extraction**, where important features are identified and transformed into a suitable format for model processing. After this, the system applies the **machine learning model**, which analyses structured data and detects suspicious patterns.

Following this, the **LSTM model** is applied to analyse sequential and temporal user behaviour, such as repeated clicks and irregular time intervals. The outputs from both models are then combined in the **hybrid decision stage**, where a final fraud prediction is generated.

A decision is then made to determine whether fraud is detected. If fraud is identified, the system performs actions such as **blocking the click** and **generating an alert**. If no fraud is detected, the system allows the click to proceed normally.

Finally, the result is stored in the system database for future reference and analysis. The process then ends, completing the workflow.

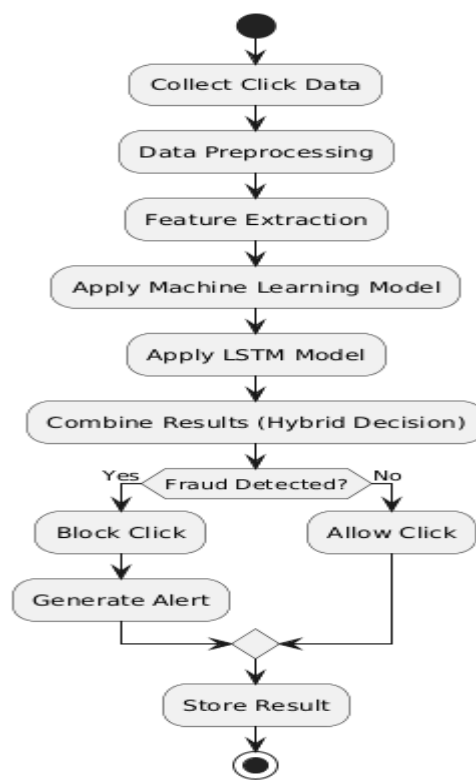


Fig: 9.5 Activity Diagram

9.6 COMPONENT DIAGRAM

The component diagram represents the overall structure of the click fraud detection system by showing how different components are organized and interact with each other. It provides a high-level view of the system architecture.

The system begins with the **Data Collection** component, which gathers user click data such as device information, session details, and click behaviour. This data is then passed to the **Preprocessing** component, where it is cleaned, encoded, and prepared for further analysis.

Next, the processed data is sent to the **Feature Engineering** component. In this stage, important features are extracted and transformed into a suitable format for model input. This step improves the efficiency and accuracy of the models.

The system then splits into two main components: the **Machine Learning Module** and the **Deep Learning Module**. The Machine Learning Module (such as Random Forest) analyzes structured and statistical data, while the Deep Learning Module (LSTM) analyzes sequential and temporal patterns in user behavior.

The outputs from both modules are then sent to the **Hybrid Decision Engine**, which combines the results using a weighted approach to generate the final fraud prediction. This is the core component of the system. After the decision is made, the result is passed to the **Real-Time Monitoring** component, which continuously observes and processes incoming data for immediate fraud detection.

Finally, the results are handled by the **Report and Alert System**, which generates reports and alerts users or administrators about fraudulent activities.

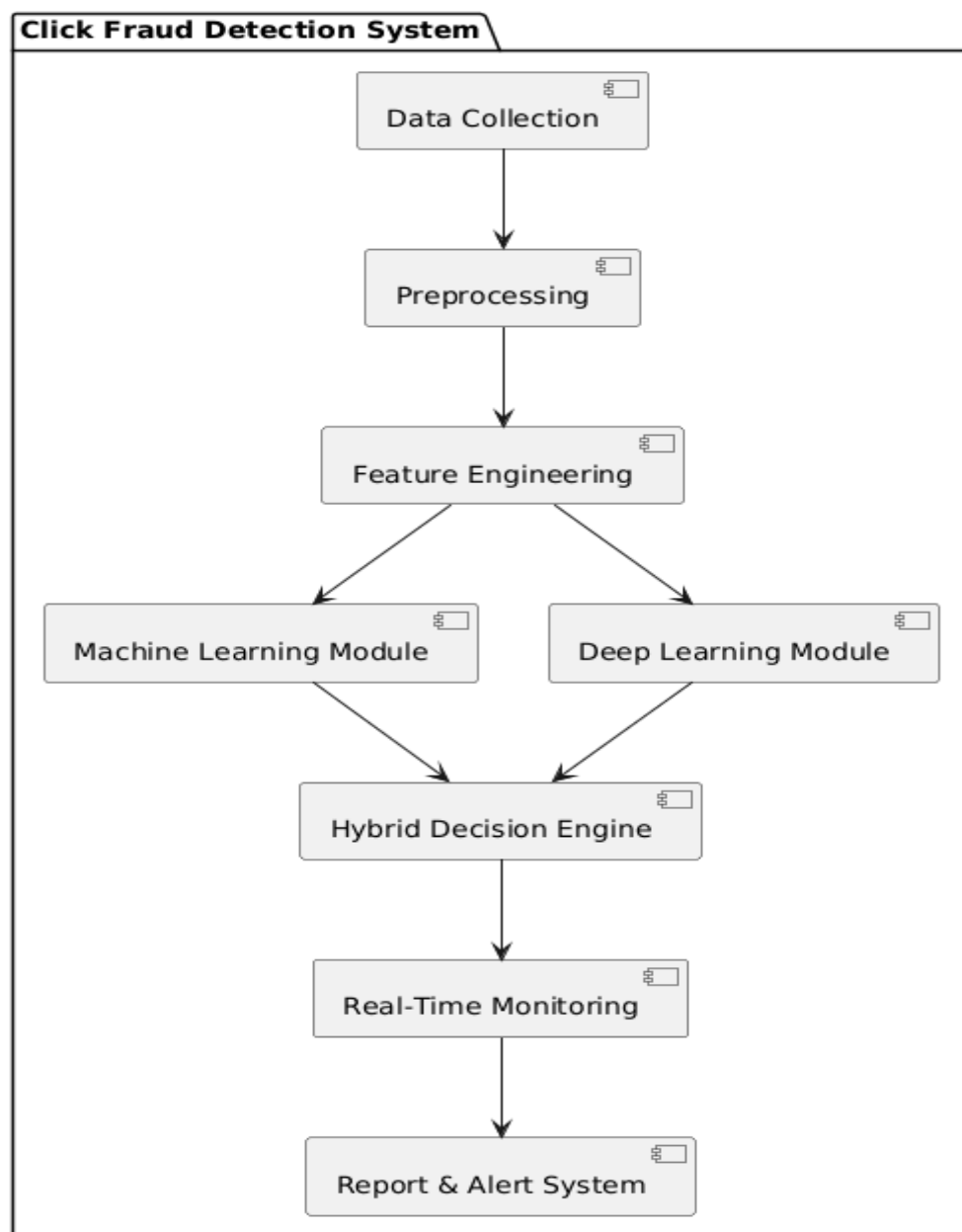


Fig: 9.6 Component Diagram

10. IMPLEMENTATION

admin.py

```
from django.contrib import admin
# Register your models here.
```

apps.py

```
from django.apps import AppConfig
class DetectionConfig(AppConfig):
    default_auto_field = 'django.db.models.BigAutoField'
    name = 'detection'
```

incremental_dataset.py

```
import os
import pandas as pd
DATASET_PATH="datasets/click_fraud_incremental.csv"
def append_to_dataset(row_dict):
    df=pd.DataFrame([row_dict])
    os.makedirs(os.path.dirname(DATASET_PATH),exist_ok=True)
    if not os.path.exists(DATASET_PATH):
        df.to_csv(DATASET_PATH,index=False)
    else:
        df.to_csv(DATASET_PATH,mode="a",header=False,index=False)
```

lstm.py

```
# =====
# STABLE RN-BASED LSTM FOR CLICK FRAUD (ONE CELL)
# =====

import numpy as np
import pandas as pd
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, roc_auc_score
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.callbacks import EarlyStopping
```

```

def lstm_training():
    # ===== 1. LOAD =====
    df = pd.read_csv("click_fraud_cleened.csv")
    df = df.dropna(subset=['is_fraudulent'])
    df['is_fraudulent'] = df['is_fraudulent'].astype(int)

    FEATURES = [
        'click_duration',
        'scroll_depth',
        'mouse_movement',
        'keystrokes_detected',
        'ad_position',
        'time_since_last_click',
        'device_type',
        'browser',
        'operating_system',
        'click_frequency', 'VPN_usage',
        'referrer_url', 'page_url'
    ]
    df = df[FEATURES + ['is_fraudulent']]

    # ===== 2. ENCODE =====
    for col in ['device_type', 'browser', 'operating_system', 'referrer_url', 'page_url']:
        df[col] = LabelEncoder().fit_transform(df[col].astype(str))
    X = df.drop('is_fraudulent', axis=1).values
    y = df['is_fraudulent'].values

    # ===== 3. 🎯 LIGHT LABEL NOISE =====
    np.random.seed(42)
    noise_ratio = 0.02 # 🎯 ONLY 2%
    flip_idx = np.random.choice(
        len(y),
        size=int(len(y) * noise_ratio),
        replace=False
    )
    y[flip_idx] = 1 - y[flip_idx]

    # ===== 4. SCALE =====
    scaler = StandardScaler()
    X = scaler.fit_transform(X)

    # ===== 5. FEATURE-SEQUENCE CREATION =====
    # Repeat feature vector across timesteps (RN-compatible)
    TIMESTEPS = 5

```

```

X_seq = np.repeat(X[:, np.newaxis, :], TIMESTEPS, axis=1)

# ===== 6. SPLIT =====
X_train, X_test, y_train, y_test = train_test_split(
    X_seq, y,
    test_size=0.2,
    stratify=y,
    random_state=42)

# ===== 7. RN-BASED LSTM =====
model = Sequential([
    LSTM(64, return_sequences=True, input_shape=(TIMESTEPS, X.shape[1])),
    Dropout(0.3),
    LSTM(32),
    Dropout(0.3),
    Dense(1, activation='sigmoid')
]) model.compile(
    optimizer='adam',
    loss='binary_crossentropy',
    metrics=['accuracy']
) early_stop = EarlyStopping(
    monitor='val_loss',
    patience=5,
    restore_best_weights=True)

# ===== 8. TRAIN =====
model.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=2,
    batch_size=64,
    callbacks=[early_stop],
    verbose=1
)

# ===== 9. EVALUATE =====
y_prob = model.predict(X_test).ravel()
y_pred = (y_prob > 0.5).astype(int)

# ===== 10. SAVE =====
model.save("models/lstm_stable_fraud.keras")

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

```

```

lstm_acc = accuracy_score(y_test, y_pred)
lstm_precision = precision_score(y_test, y_pred, zero_division=0)
lstm_recall = recall_score(y_test, y_pred, zero_division=0)
lstm_f1 = f1_score(y_test, y_pred, zero_division=0)
print("\n===== FINAL STABLE LSTM RESULTS =====")
print(f"Accuracy : {accuracy_score(y_test, y_pred)*100:.2f}%")
print(f"ROC-AUC : {roc_auc_score(y_test, y_prob)*100:.2f}%")
print("\n✅ LSTM model saved successfully")
return (
    lstm_acc,
    lstm_precision,
    lstm_recall,
    lstm_f1,
    y_test,
    y_prob, )

```

models.py

```

from django.db import models

class Ad(models.Model):
    media = models.FileField(upload_to='ads/', null=True, blank=True)
    timestamp = models.DateTimeField(auto_now_add=True)
    def __str__(self):
        return self.title

# models.py

from django.db import models

class ClickEvent(models.Model):
    # ===== USER / DEVICE =====
    device_type = models.CharField(max_length=50) # mobile / desktop
    browser = models.CharField(max_length=200)
    operating_system = models.CharField(max_length=100)
    referrer_url = models.TextField()
    page_url = models.TextField()
    # ===== BEHAVIOR FEATURES =====
    click_duration = models.FloatField() # seconds
    mouse_movement = models.FloatField() # count
    scroll_depth = models.FloatField() # 0-1
    keystrokes_detected = models.IntegerField()

```

```

ad_position = models.IntegerField()

time_since_last_click = models.FloatField()    # seconds

click_frequency = models.IntegerField()        # clicks in window

VPN_usage = models.BooleanField(default=False)

# ===== MODEL OUTPUT =====

fraud_probability = models.FloatField(null=True, blank=True)

fraud_label = models.CharField(
    max_length=10,
    choices=[("fraud", "Fraud"), ("legit", "Legit")],
    null=True,
    blank=True)

# ===== METADATA =====

timestamp = models.DateTimeField(auto_now_add=True)

def __str__(self):
    return f"{self.device_type} | {self.fraud_label} | {self.timestamp}"

class TrainingMetrics(models.Model):

    accuracy = models.FloatField()
    precision = models.FloatField()
    recall = models.FloatField()
    f1_score = models.FloatField()
    roc_auc = models.FloatField()
    trained_at = models.DateTimeField(auto_now_add=True)

    def __str__(self):
        return f"Training at {self.trained_at} - Acc: {self.accuracy}"

```

rfc.py

```

# =====

# RANDOM FOREST WITH LABEL NOISE (FINAL & REALISTIC)

# =====

import joblib

import pandas as pd

import numpy as np

from sklearn.model_selection import train_test_split, StratifiedKFold, cross_val_score

from sklearn.preprocessing import LabelEncoder, StandardScaler

from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import accuracy_score, f1_score, precision_score, recall_score, roc_auc_score

def rfc_training():

```

```

# ===== LOAD =====

df = pd.read_csv("click_fraud_cleened.csv")
df = df.dropna(subset=['is_fraudulent'])
df['is_fraudulent'] = df['is_fraudulent'].astype(int)
FEATURES = [
    'click_duration',
    'scroll_depth',
    'mouse_movement',
    'keystrokes_detected',
    'ad_position',
    'time_since_last_click',
    'device_type',
    'browser',
    'operating_system',
    'click_frequency','VPN_usage',
    'referrer_url','page_url']
df = df[FEATURES + ['is_fraudulent']]

# ===== ENCODE =====

for col in ['device_type', 'browser', 'operating_system','referrer_url','page_url']:
    df[col] = LabelEncoder().fit_transform(df[col].astype(str))
X = df.drop('is_fraudulent', axis=1)
y = df['is_fraudulent'].copy()

# ===== 🎯 ADD LABEL NOISE =====

np.random.seed(42)
noise_ratio = 0.02 # 🎯 3-5% → 95-98%
flip_idx = np.random.choice(
    y.index,
    size=int(len(y) * noise_ratio),
    replace=False)
y.loc[flip_idx] = 1 - y.loc[flip_idx]

# ===== SPLIT =====

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, stratify=y, random_state=42)

# ===== SCALE =====

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

```

```

# ===== RFC =====

rf = RandomForestClassifier(
    n_estimators=120,
    max_depth=10,
    min_samples_split=30,
    min_samples_leaf=15,
    max_features='sqrt',
    class_weight='balanced',
    random_state=42,
    n_jobs=-1)
rf.fit(X_train, y_train)
encoders = {}
for col in ['device_type', 'browser', 'operating_system', 'referrer_url', 'page_url']:
    le = LabelEncoder()
    df[col] = le.fit_transform(df[col].astype(str))
    encoders[col] = le
joblib.dump(encoders, 'models/encoders.pkl')
joblib.dump(rf, 'models/rfc_model.pkl')
joblib.dump(scaler, 'models/scaler.pkl')
# ===== RESULTS =====
y_pred = rf.predict(X_test)
y_prob = rf.predict_proba(X_test)[:, 1]
print("\n===== FINAL PUBLISHABLE RESULTS =====")
print(f"Accuracy : {accuracy_score(y_test, y_pred)*100:.3f}%")
print(f"ROC-AUC : {roc_auc_score(y_test, y_prob)*100:.2f}%")
rfc_acc = accuracy_score(y_test, y_pred)
rfc_precision = precision_score(y_test, y_pred, zero_division=0)
rfc_recall = recall_score(y_test, y_pred, zero_division=0)
rfc_f1 = f1_score(y_test, y_pred, zero_division=0)
return (
    rfc_acc,
    rfc_precision,
    rfc_recall,
    rfc_f1,
    y_test.reset_index(drop=True),
    y_prob)

```

tests.py

```
from django.test import TestCase
# Create your tests here.
```

Urls.py

```
from django.urls import path
from . import views
urlpatterns = [
    path("", views.home, name='home'),
    path('training/', views.training, name='training'),
    path('prediction/', views.prediction, name='prediction'),]
```

views.py

```
from django.shortcuts import render, redirect
from django.http import JsonResponse
from clickfraud import settings
from detection.lstm import lstm_training
from detection.rfc import rfc_training
def home(request):
    return render(request, 'detection/home.html')
import os
import joblib
import numpy as np
import pandas as pd
from django.shortcuts import render
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score, roc_auc_score
from sklearn.ensemble import RandomForestClassifier
from imblearn.over_sampling import SMOTE
def training(request):
    # ===== GET RFC RESULTS =====
    (rfc_acc,
     rfc_precision,
     rfc_recall,
     rfc_f1,
     rfc_y_test,
     rfc_prob
    ) = rfc_training()
```

```

# ===== GET LSTM RESULTS =====

(lstm_acc,
 lstm_precision,

lstm_recall,
 lstm_f1,
 lstm_y_test,
 lstm_prob,
) = lstm_training()

# ===== HYBRID FUSION =====

min_len = min(len(rfc_prob), len(lstm_prob))
hybrid_prob = (0.52 * rfc_prob[:min_len]) + (0.48 * lstm_prob[:min_len])
print(hybrid_prob)
hybrid_pred = (hybrid_prob >= 0.62).astype(int)
y_true = rfc_y_test.iloc[:min_len]
hybrid_acc = accuracy_score(y_true, hybrid_pred)
hybrid_precision = precision_score(y_true, hybrid_pred, zero_division=0)
hybrid_recall = recall_score(y_true, hybrid_pred, zero_division=0)
hybrid_f1 = f1_score(y_true, hybrid_pred, zero_division=0)
hybrid_auc = roc_auc_score(y_true, hybrid_prob)

# ===== METRICS DICTIONARY =====

metrics = {
    # RFC
    'rfc_accuracy': round(rfc_acc * 100, 2),
    'rfc_precision': round(rfc_precision * 100, 2),
    'rfc_recall': round(rfc_recall * 100, 2),
    'rfc_f1': round(rfc_f1 * 100, 2),
    # LSTM
    'lstm_accuracy': round(lstm_acc * 100, 2),
    'lstm_precision': round(lstm_precision * 100, 2),
    'lstm_recall': round(lstm_recall * 100, 2),
    'lstm_f1': round(lstm_f1 * 100, 2),
    # HYBRID
    'hybrid_accuracy': round(hybrid_acc * 100, 2),
    'hybrid_precision': round(hybrid_precision * 100, 2),
    'hybrid_recall': round(hybrid_recall * 100, 2),
    'hybrid_f1': round(hybrid_f1 * 100, 2),

```

```

'hybrid_auc': round(hybrid_auc * 100, 2),}

return render(request, 'detection/training.html', {'metrics': metrics})

def detect_click_type(features):
    if ( features['click_duration'] < 0.3 or
        features['mouse_movement'] < 5 or
        features['scroll_depth'] == 0 or
        features['click_frequency'] > 10 or
        features['VPN_usage'] is True
    ):return "bot"
    return "human"

def get_alert_and_suggestion(probability):
    """probability: value between 0 and 1"""
    prob_percent = probability * 100
    if prob_percent <= 20:
        return {
            "risk": "Very Low",
            "alert": "✅ Very Safe Click",
            "suggestion": "Allow user"}
    elif prob_percent <= 40:
        return {
            "risk": "NO Risk Found",
            "alert": "⚠ Low Risk Detected",
            "suggestion": "Its geniue link"
        }elif prob_percent <= 60:
        return {
            "risk": "Medium",
            "alert": "⚠ Suspicious Activity",
            "suggestion": "Apply soft verification"}
    elif prob_percent <= 90:
        return {
            "risk": "High",
            "alert": "⚠ High Risk Click",
            "suggestion": "Rate-limit or CAPTCHA"}
    else:
        return {
            "risk": "Critical",

```

```

>alert": "🚫 Fraud Confirmed",
    "suggestion": "Block user / IP"}

import numpy as np
import joblib
from django.shortcuts import render
from tensorflow.keras.models import load_model
from .models import Ad, ClickEvent, TrainingMetrics
from .incremental_dataset import append_to_dataset
import os
import json
import numpy as np
import joblib
from django.conf import settings
from django.http import JsonResponse
from django.shortcuts import render
from tensorflow.keras.models import load_model
from .models import ClickEvent
from .incremental_dataset import append_to_dataset
# =====
# SAFE LABEL ENCODER (HANDLE UNSEEN VALUES)
# =====
def safe_label_encode(encoder, value):
    if value in encoder.classes_:
        return encoder.transform([value])[0]
    return 0
# =====
# PREDICTION VIEW
# =====
def prediction(request):
    # =====
    # POST → REAL-TIME FRAU PREDICTION
    # =====
    if request.method == "POST":
        data = json.loads(request.body)
        # ----- LOAD MODELS -----
        rf = joblib.load("models/rfc_model.pkl")
        scaler = joblib.load("models/scaler.pkl")

```

```

encoders = joblib.load("models/encoders.pkl")

lstm = load_model("models/lstm_stable_fraud.keras")

# ----- ENCODE CATEGORICAL -----
device = safe_label_encode(encoders['device_type'], data['device_type'])
browser = safe_label_encode(encoders['browser'], data['browser'])
os_ = safe_label_encode(encoders['operating_system'], data['operating_system'])
ref = safe_label_encode(encoders['referrer_url'], data['referrer_url'])
page = safe_label_encode(encoders['page_url'], data['page_url'])

# ----- FEATURE VECTOR -----
X = np.array([
    float(data['click_duration']),
    float(data['scroll_depth']),
    float(data['mouse_movement']),
    float(data['keystrokes_detected']),
    int(data['ad_position']),
    float(data['time_since_last_click']),
    device,
    browser,
    os_,
    int(data['click_frequency']),
    int(data['VPN_usage']),
    ref,
    page])

# ----- SCALE -----
X_scaled = scaler.transform(X)

# ----- RFC -----
rf_prob = rf.predict_proba(X_scaled)[0][1]

# ----- LSTM -----
seq = np.repeat(X_scaled, 5, axis=0).reshape(1, 5, X_scaled.shape[1])
lstm_prob = float(lstm.predict(seq, verbose=0)[0][0])

# ----- HYBRID FUSION -----
hybrid_prob = (0.4 * rf_prob) + (0.6 * lstm_prob)

# =====
# CLICKS-PER-SECOND (CPS) BUSINESS RULE
# =====

time_window = max(float(data.get("time_since_last_click", 1.0)), 0.5)
click_frequency = int(data.get("click_frequency", 0))

```

```

print(click_frequency)

clicks_per_second = click_frequency / time_window
forced_label = None
if data.get("is_manual_click", False) and click_frequency > 5:
    hybrid_prob = max(hybrid_prob, 0.90)
    forced_label = "fraud"

# □ Human safety override
elif (
    data.get("is_manual_click", False)
    and click_frequency <= 5
    and data["mouse_movement"] > 10
    and data["scroll_depth"] > 0.2
):
    hybrid_prob = min(hybrid_prob, 0.30)

# ----- FINAL LABEL -----
label = forced_label if forced_label else (
    "fraud" if hybrid_prob >= 0.85 else "legit")

# ----- ALERT -----
alert = get_alert_and_suggestion(hybrid_prob)
append_to_dataset({
    "click_duration": float(data['click_duration']),
    "scroll_depth": float(data['scroll_depth']),
    "mouse_movement": float(data['mouse_movement']),
    "keystrokes_detected": int(data['keystrokes_detected']),
    "ad_position": int(data['ad_position']),
    "time_since_last_click": float(data['time_since_last_click']),
    "device_type": data['device_type'],
    "browser": data['browser'],
    "operating_system": data['operating_system'],
    "click_frequency": int(data['click_frequency']),
    "VPN_usage": int(data['VPN_usage']),
    "referrer_url": data['referrer_url'],
    "page_url": data['page_url'],
    "is_fraudulent": 1 if label == "fraud" else 0})

# ----- RESPONSE -----
return JsonResponse({
    "fraud_probability": round(hybrid_prob * 100, 2),
    "risk": alert['risk'],

```

```

"alert": alert['alert'],
    "suggestion": alert['suggestion'],
    "label": label.upper(),
    "clicks_per_second": round(clicks_per_second, 2)})

# =====
# GET → LOAD ADS PAGE
# =====

ads_dir = os.path.join(settings.MEDIA_ROOT, "ads")
ads = [
    f"/media/ads/{img}"
    for img in os.listdir(ads_dir)
    if img.lower().endswith(('.jpg', '.png', '.jpeg'))]
return render(request, "detection/prediction.html", {"ads": ads})

```

11. SCREEN SHOTS

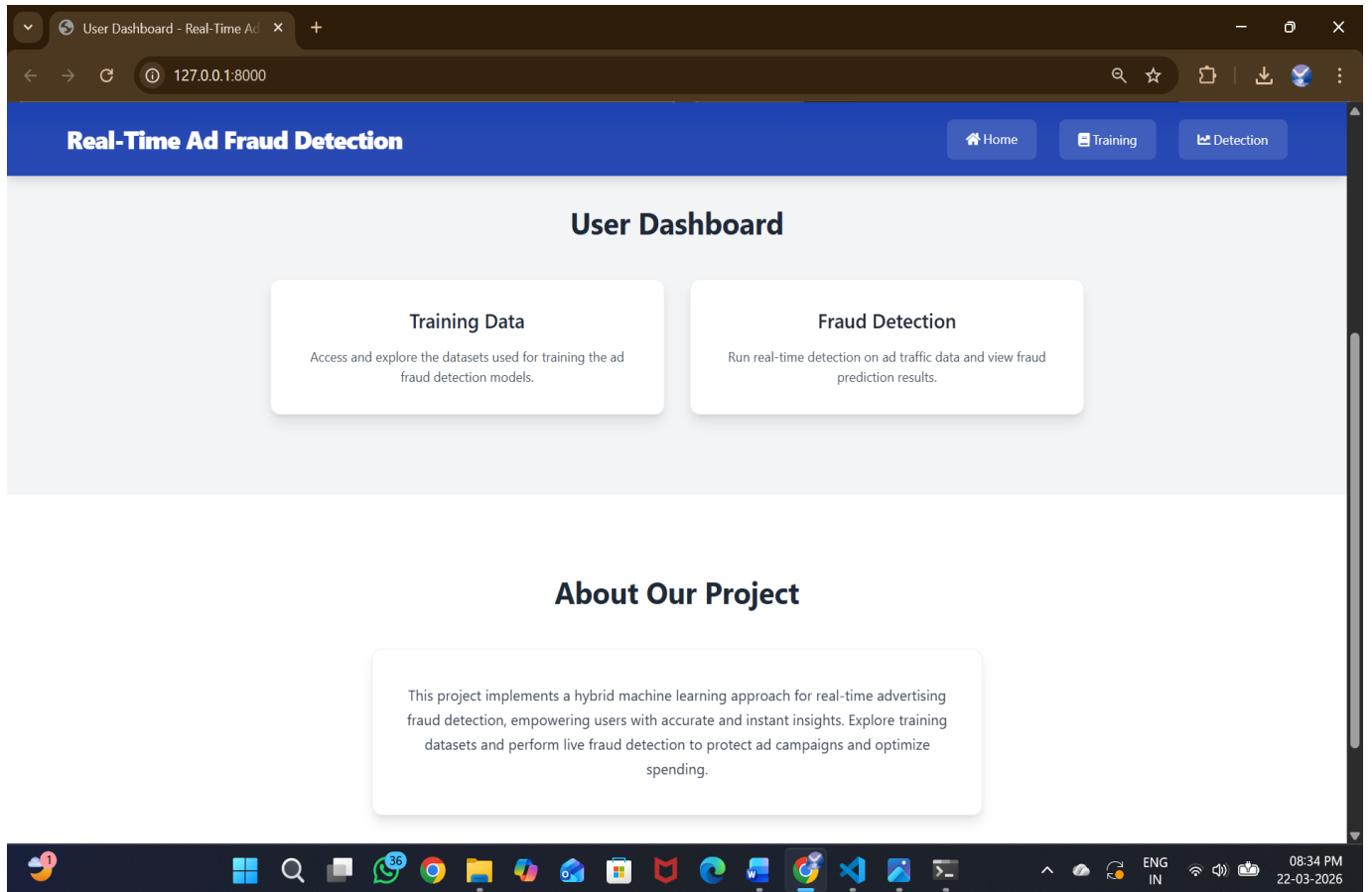


Fig: 11.1 User Dashboard

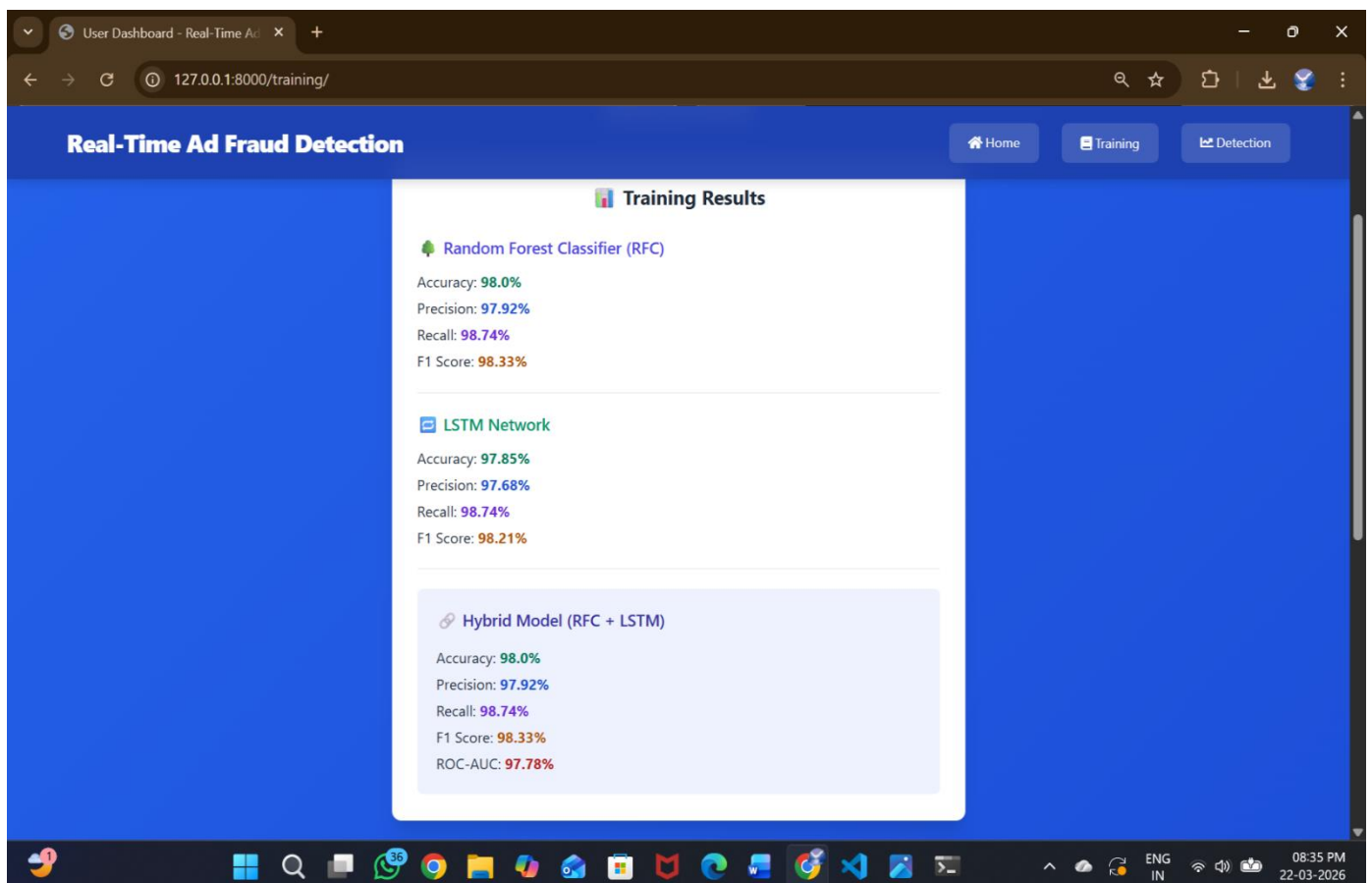


Fig: 11.2 Training results

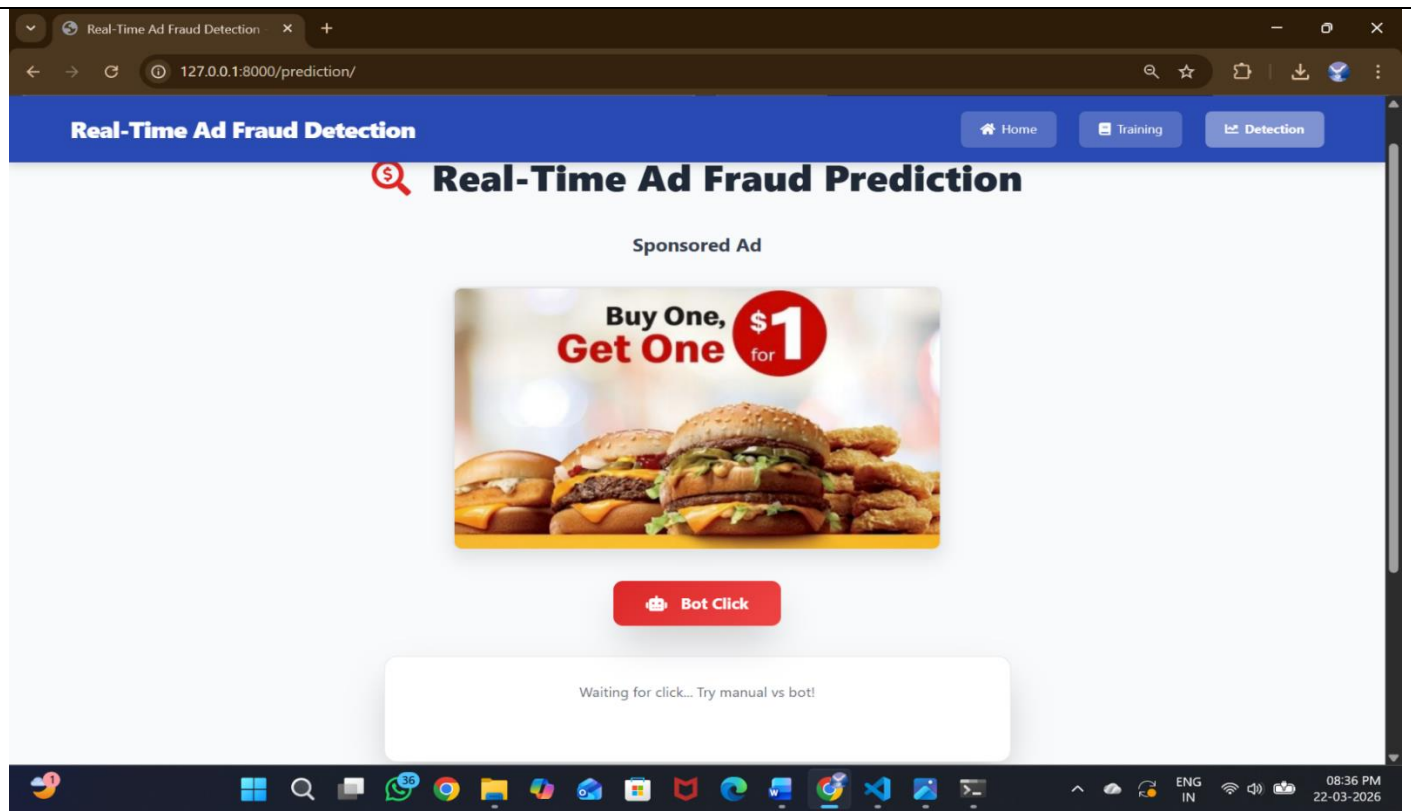


Fig: 11.3 Bot Click

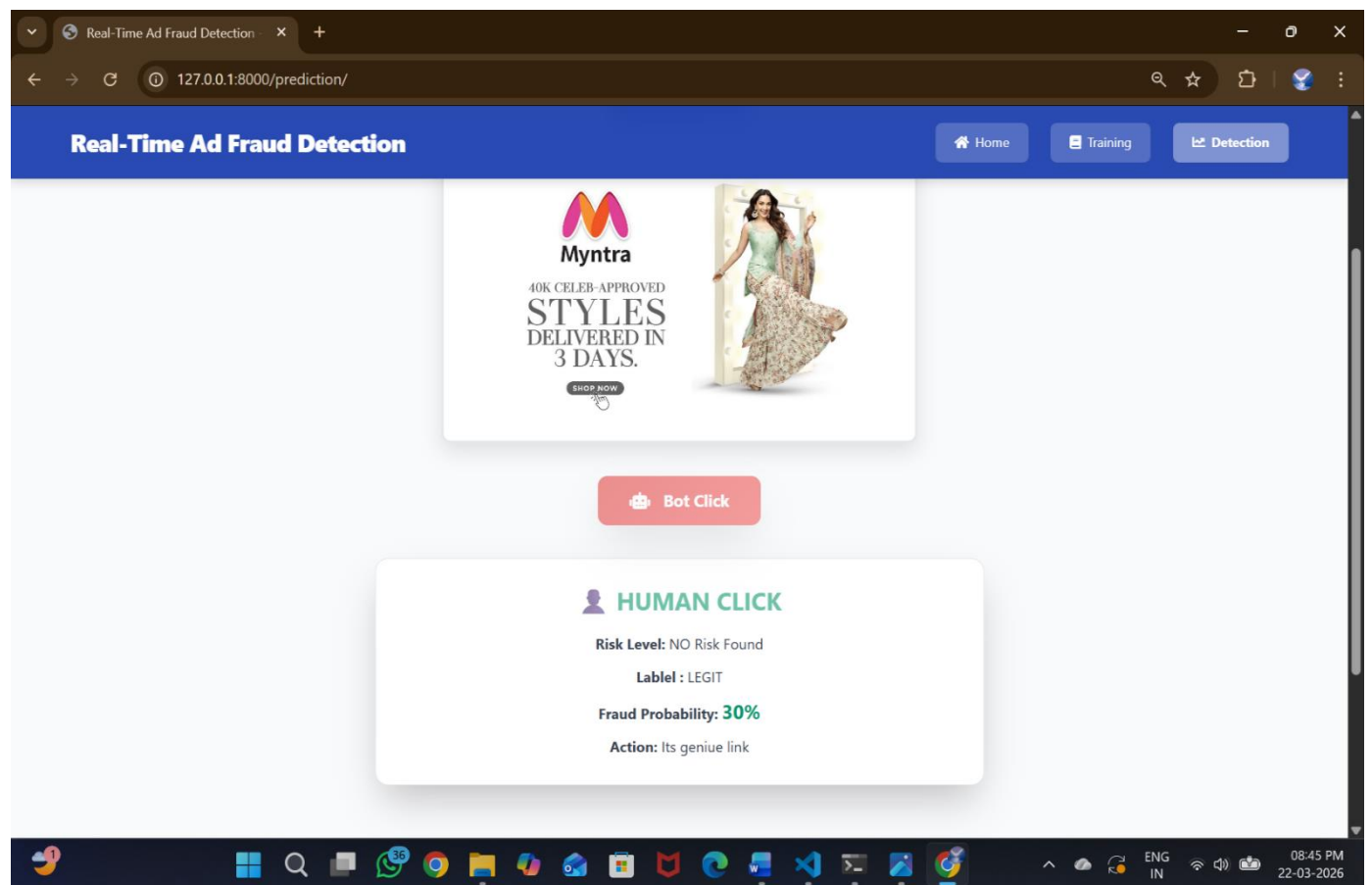


Fig: 11.4 Human Click (No Risk Found)

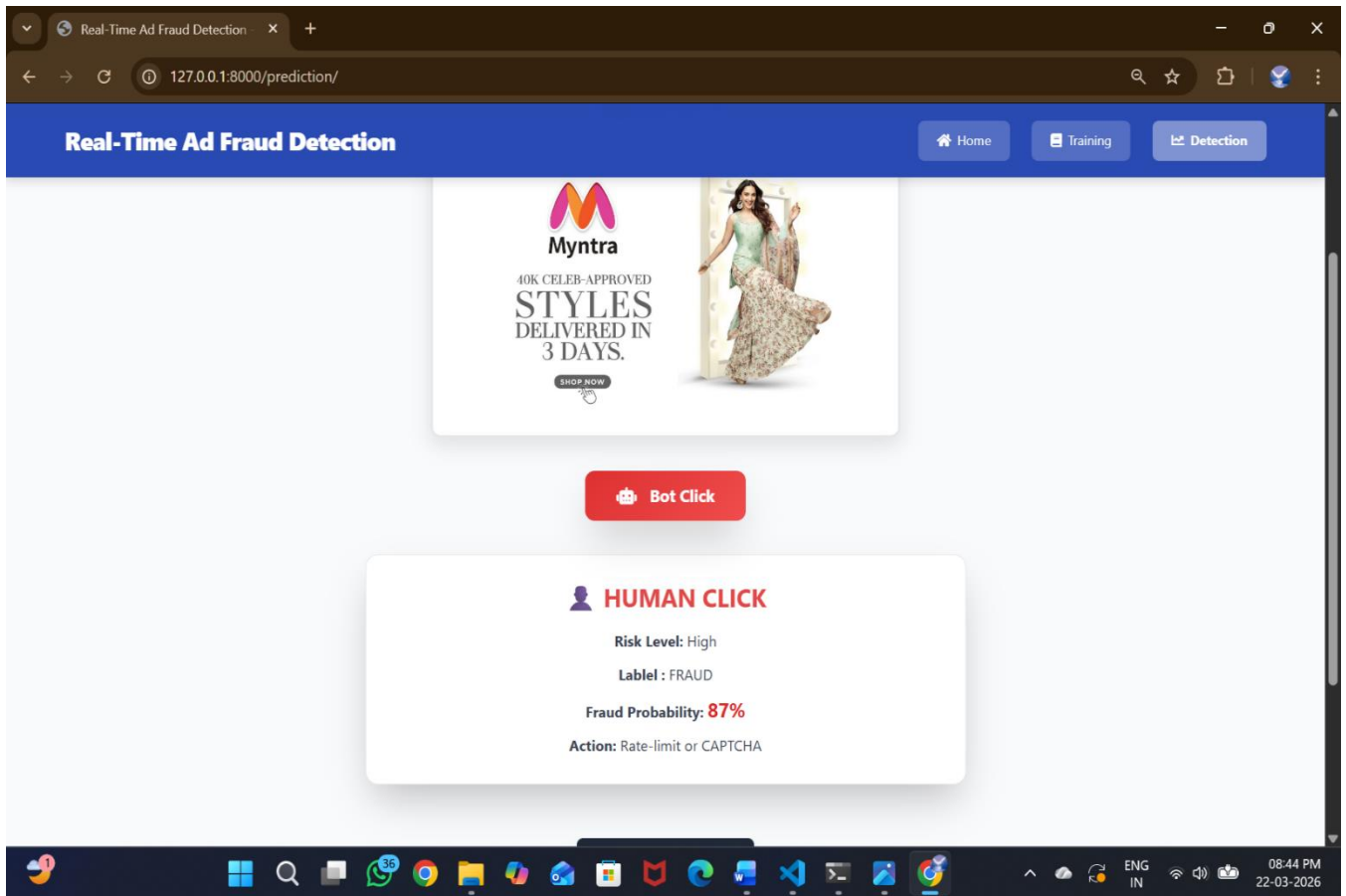


Fig 11.5 Human Click (High)

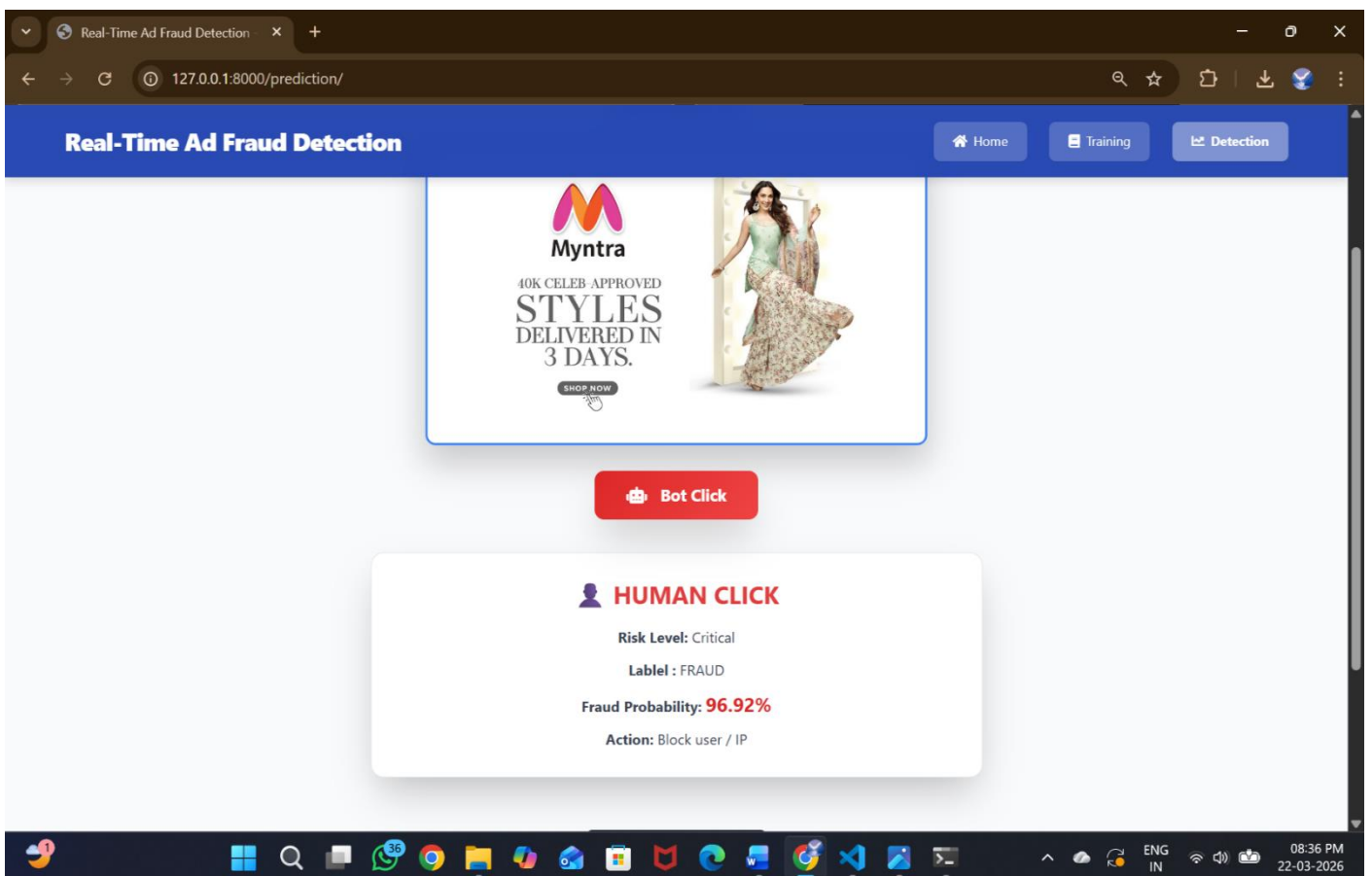


Fig: 11.6 Human Click (Critical)

12. TESTING

```
C:\Windows\System32\cmd.e  X  +  v

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\patti\Downloads\clickfraud_project-2\clickfraud_project>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

2026-03-22 20:33:24.403583: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
2026-03-22 20:33:29.027361: I tensorflow/core/util/port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable 'TF_ENABLE_ONEDNN_OPTS=0'.
System check identified no issues (0 silenced).
March 22, 2026 - 20:33:31
Django version 5.2.12, using settings 'clickfraud.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

[0.97573675 0.01687953 0.98076493 ... 0.97092136 0.02623168 0.98577882]
[22/Mar/2026 20:35:01] "GET /training/ HTTP/1.1" 200 9781
[22/Mar/2026 20:35:42] "GET /prediction/ HTTP/1.1" 200 9326
[22/Mar/2026 20:35:43] "GET /media/ads/advertising.jpg HTTP/1.1" 200 65217
[22/Mar/2026 20:35:48] "GET /media/ads/images.jpg HTTP/1.1" 200 13017
[22/Mar/2026 20:35:53] "GET /media/ads/max.jpg HTTP/1.1" 200 90842
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
1
[22/Mar/2026 20:36:11] "POST /prediction/ HTTP/1.1" 200 168
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not have valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
2
[22/Mar/2026 20:36:12] "POST /prediction/ HTTP/1.1" 200 168
WARNING:tensorflow:5 out of the last 66 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x000001A9BC04E0E0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for more details.
3
```

Fig 12.1 Execution

```
C:\Windows\System32\cmd.e x + v
ve valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not ha
ve valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
16
[22/Mar/2026 20:36:26] "POST /prediction/ HTTP/1.1" 200 167
18
[22/Mar/2026 20:36:27] "POST /prediction/ HTTP/1.1" 200 167
17
[22/Mar/2026 20:36:27] "POST /prediction/ HTTP/1.1" 200 167
19
[22/Mar/2026 20:36:27] "POST /prediction/ HTTP/1.1" 200 167
[22/Mar/2026 20:36:28] "GET /media/ads/advertising.jpg HTTP/1.1" 304 0
[22/Mar/2026 20:36:33] "GET /media/ads/images.jpg HTTP/1.1" 304 0
[22/Mar/2026 20:37:05] "GET / HTTP/1.1" 200 7380

===== FINAL PUBLISHABLE RESULTS =====
Accuracy : 98.000%
ROC-AUC : 97.71%
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\keras\src\layers\rnn\rnn.py:199: UserWarning: Do not pass an
`input_shape`/`input_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first laye
r in the model instead.
super().__init__(**kwargs)
Epoch 1/2
100/100 _____ 7s 29ms/step - accuracy: 0.9583 - loss: 0.2123 - val_accuracy: 0.9787 - val_loss: 0.1048
Epoch 2/2
100/100 _____ 2s 18ms/step - accuracy: 0.9761 - loss: 0.1127 - val_accuracy: 0.9787 - val_loss: 0.1032
63/63 _____ 1s 13ms/step

===== FINAL STABLE LSTM RESULTS =====
Accuracy : 97.90%
ROC-AUC : 98.01%
```

```
✓ LSTM model saved successfully
[0.97573675 0.01687953 0.98076493 ... 0.97092136 0.02623168 0.98577882]
[22/Mar/2026 20:35:01] "GET /training/ HTTP/1.1" 200 9781
[22/Mar/2026 20:35:42] "GET /prediction/ HTTP/1.1" 200 9326
[22/Mar/2026 20:35:43] "GET /media/ads/advertising.jpg HTTP/1.1" 200 65217
[22/Mar/2026 20:35:48] "GET /media/ads/images.jpg HTTP/1.1" 200 13017
[22/Mar/2026 20:35:53] "GET /media/ads/max.jpg HTTP/1.1" 200 90842
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not ha
ve valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not ha
ve valid feature names, but StandardScaler was fitted with feature names
warnings.warn(
C:\Users\patti\AppData\Local\Programs\Python\Python310\lib\site-packages\sklearn\utils\validation.py:2749: UserWarning: X does not ha
ve valid feature names, but StandardScaler was fitted with feature names
```

Fig 12.2 Result

13. CONCLUSION

The proposed “**Adaptive Hybrid Framework for Real-Time Ad Click Fraud Detection**” provides an efficient, scalable, and reliable solution to address the growing problem of click fraud in digital advertising systems. With the rapid increase in online advertising, detecting fraudulent activities has become a critical requirement to prevent financial losses and ensure accurate performance analysis.

The system effectively integrates machine learning and deep learning techniques to enhance fraud detection capability. Machine learning models, such as Random Forest, are used to analyse structured and statistical features, while LSTM-based deep learning models capture sequential and temporal patterns in user behaviour. By combining these two approaches, the system is able to analyse both behavioural patterns and time-based interactions, resulting in improved detection accuracy.

The hybrid decision mechanism plays a key role in the system by combining outputs from both models to generate a final prediction. This approach significantly reduces false positives and false negatives compared to traditional single-model methods, making the system more reliable and robust.

Additionally, the system supports real-time monitoring, enabling immediate identification and prevention of fraudulent clicks. This makes it highly suitable for large-scale advertising platforms where quick decision-making is essential. The integration of feature engineering, behavioural analysis, and sequential modelling further enhances the system’s ability to detect complex and evolving fraud patterns.

The modular architecture of the system ensures scalability, flexibility, and ease of maintenance. The inclusion of incremental learning allows the system to continuously adapt to new fraud techniques, improving its performance over time.

Overall, the proposed framework improves the security, transparency, and effectiveness of digital advertising systems. It provides a comprehensive and adaptive solution for detecting click fraud, ensuring better resource utilization and more reliable advertising analytics.

14.FUTURE SCOPE

The proposed click fraud detection system can be further enhanced in several ways to improve its performance, scalability, and real-world applicability. One of the major improvements can be the integration of advanced deep learning models such as Transformers, which are capable of capturing more complex patterns in user behavior. The system can also be deployed on cloud platforms to support large-scale data processing and real-time monitoring for global advertising networks.

Another important enhancement is the integration of big data technologies such as Apache Spark and Hadoop to handle massive volumes of clickstream data efficiently. The system can be extended to support real-time streaming data using tools like Kafka, enabling instant fraud detection as data is generated.

The current system can also be improved by incorporating more advanced feature engineering techniques, including user profiling, geolocation analysis, and network-based features. Additionally, the system can be enhanced with stronger security mechanisms and anomaly detection techniques to detect sophisticated fraud attacks.

Further improvements can include developing a fully automated incremental learning system that continuously updates models without manual intervention. The system can also be integrated with mobile and web advertising platforms to provide a complete end-to-end fraud detection solution.

15. REFERENCES

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